1. This question paper contains a total of **1** page (**1** side of paper).
2. Write your name, roll number, department, and section on every side of every sheet of this booklet
3. Write final answers neatly with a blue/black pen in the given boxes.
4. **Answers written outside the box will NOT be graded.**

Total **10 Marks**

Q. 1: Write the output of the following program in the appropriate box and answer the question. (2+2+2 = 6 Marks)

```
#include<stdio.h>
void func1(int *arr){
  for(int i=1; i<=3;i++)
    (i-1)[arr] = (i%3);
}
void func2 (int a[1] , int arr){
  a[0] = arr + 1;
}
int main(){
  int arr[4] = {0,0,0,0};
  func1(arr+1);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  printf("\n");
  func2(arr, *arr);
  for(int i=0; i<4; i++)
        printf("%d ", i[arr] );
  return 0;
}
```

arr[i-1]

Output
0 1 2 0
1 1 2 0

Will the answer change if we write func2 as:

```
void func2 (int a[1] , int arr){
  int temp = arr + 1;
  a = &temp;
}
```

Explain your answer

Yes, now the array variable a that was storing address of arr[0]; would give address of temp variable. that can be different.

Q. 2: Write the output of the following sequence of instructions **long arr2[100]; int arr1[100], a1=(&arr2[50]), a2=(&arr2[10]), b1=(&arr1[50]), b2=(&arr1[10]); printf("%d %d", a1-a2 , b1-b2);** Explain your answer. (2+2 Marks)

320 160 | a1 rep. address of arr2[50]; so, a1-a2 is distance b/w arr2[50] and arr2[10]; since variable type is long; thus 40 × sizeof(long) (8).

Similarly b1-b2 would be 40 × sizeof(int) = 40 × 4 = 160.